NEWHAVEN DISPLAY
INTERNATIONAL

Product Specification

NHD-C12864WO-B1TFH#-M

Chip-On-GLass Liquid Crystal Display Module

NHD-	Newhaven Display
C12864-	128x64 Pixels
WO-	Display Type: COG
B1-	Model
T-	White LED Backlight
F-	FSTN (+)
H-	Transflective, 6:00 Optimal View, Wide Temperature
M-	Mounting Holes

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REACH Compliant

RoHS Compliant

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Table of Contents

Document Revision History 2
Mechanical Drawing 3
Pin Description 4
Wiring Diagram 4
Electrical Characteristics 5
Optical Characteristics 5
Controller Information 5
Timing Characteristics 6
Table of Commands 9
Quality Information 10

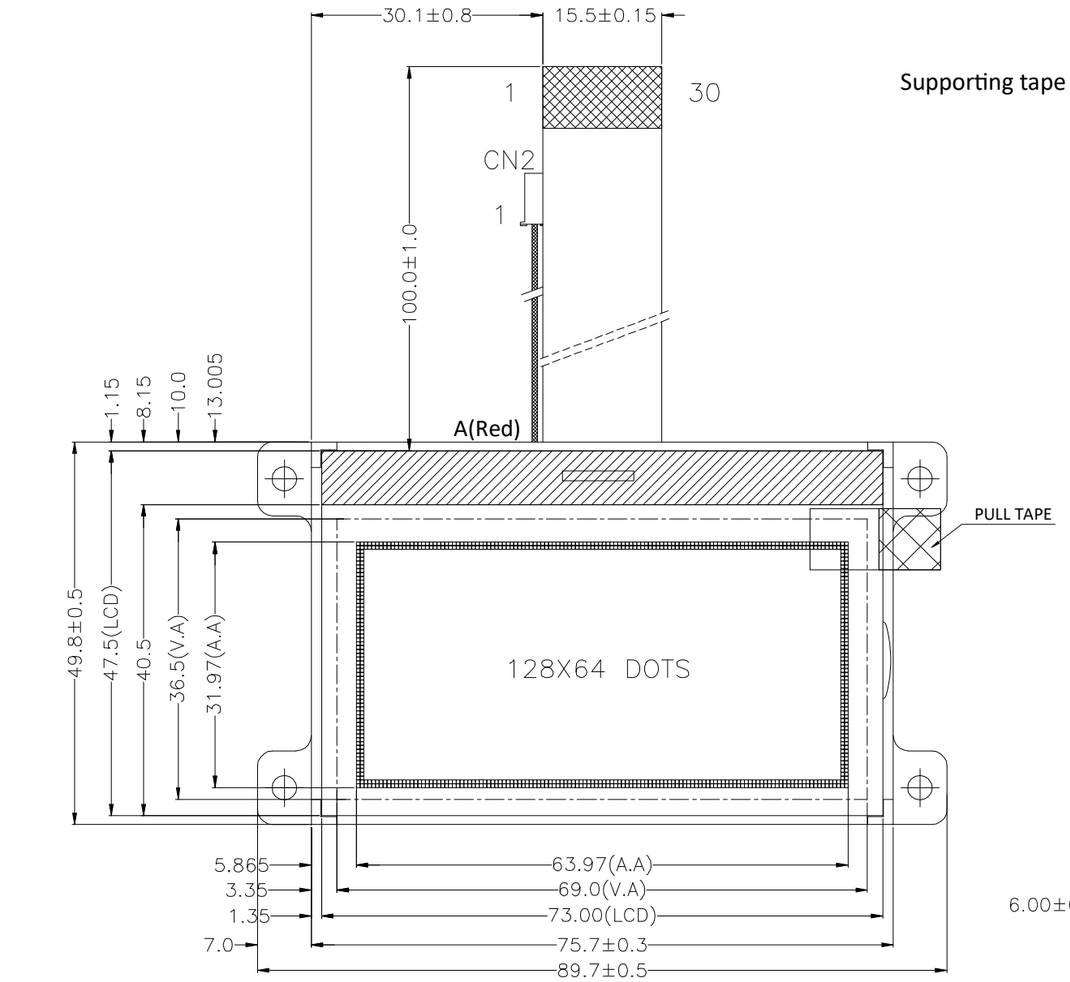
Additional Resources

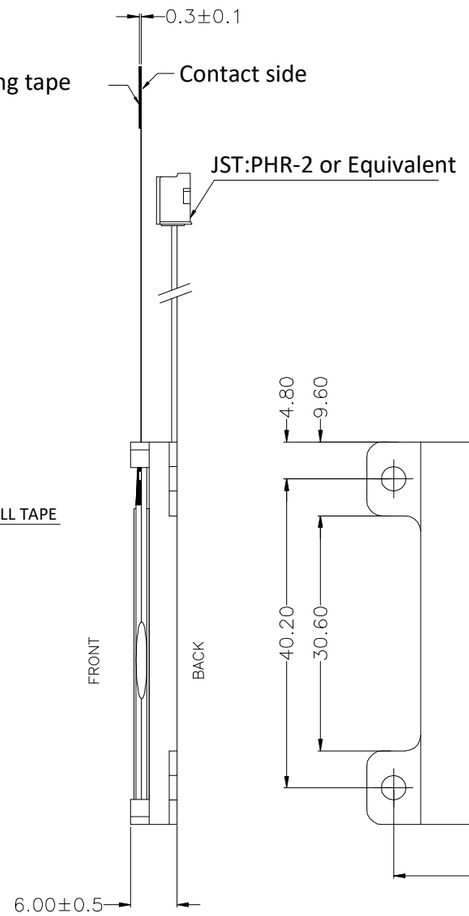
- **Support Forum:** https://support.newhavendisplay.com/hc/en-us/community/topics
- **GitHub:** https://github.com/newhavendisplay
- **Example Code:** https://support.newhavendisplay.com/hc/en-us/categories/4409527834135-Example-Code/
- **Knowledge Center:** https://www.newhavendisplay.com/knowledge_center.html
- **Quality Center:** https://www.newhavendisplay.com/quality_center.html
- **Precautions for using LCDs/LCMs:** https://www.newhavendisplay.com/specs/precautions.pdf
- **Warranty / Terms & Conditions:** https://www.newhavendisplay.com/terms.html

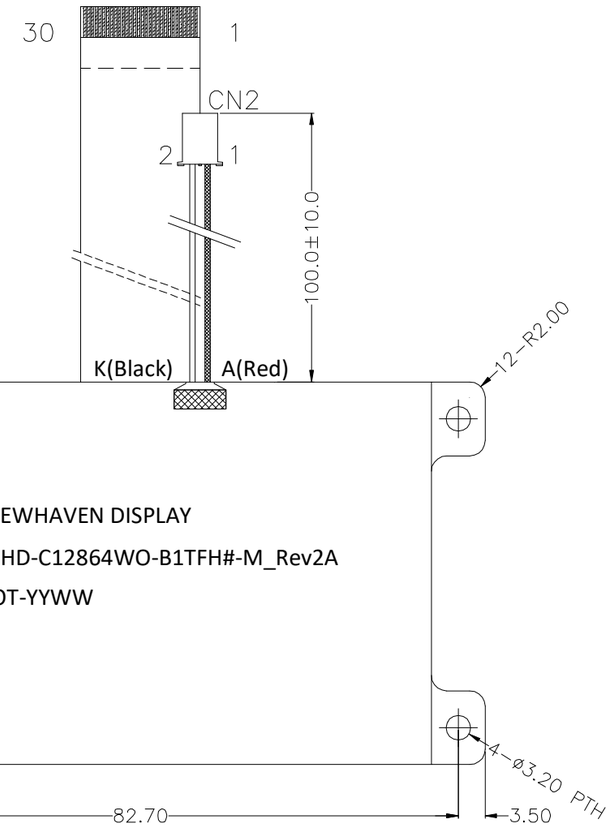
Document Revision History

Revision	Date	Description	Changed By
0	09/01/2008	Initial Release	-
1	11/12/2009	User Guide Reformat	MC
2	12/14/2009	Block Diagram Update	MC
3	02/12/2010	Added –M to p/n to signify this part has mounting holes	MC
4	03/09/2010	Block Diagram Updated	BE
5	05/04/2010	Block Diagram and Pin Description	MP
6	05/05/2010	Backlight Connectors Updated	BE
7	05/14/2010	Pin Description Update	MP
8	05/02/2011	Example Initialization Code Updated	AK
9	05/13/2011	Block Diagram Updated	AK
10	05/19/2011	Block Diagram & Pin Description Updated	AK
11	01/10/2012	Pin Description & Controller Link Updated	AK
12	06/15/2012	Timing Characteristics Updated	AK
13	04/04/2013	Backlight Mating Connector, LCD Voltage Levels Updated	AK
14	04/15/2013	Optical Characteristics Updated	AK
15	03/16/2015	Pin Description Updated	RM
16	12/30/2016	Electrical Characteristics & Response Time Updated	SB
17	02/06/2017	ILED Updated	SB
18	04/16/2020	Initialization Code Updated	TM
19	03/03/2021	Updated Tolerance on 2D Mechanical Drawing, MIN Supply Current, Backlight Drive Conditions & Quality Information	AS
20	01/11/2022	Controller IC changed from ST7565P to ST7567, Pin Description Updated, Part Revision Upgraded to Rev1A	ZP
21	04/04/2022	Backlight info and Drawing Label Updated	ZP
22	05/04/2022	Corrected Static Electricity Typo “kΩ -> Ω”	ZP
23	01/04/2023	Mechanical Drawing Updated	KL
24	04/19/2023	Updated Backlight Driven Condition	KL
25	05/11/2023	Part Revision Updated to Rev2A	KL
26	07/26/2023	Mechanical Drawing, Electrical Characteristics and Quality Information Updated	KL
27	02/29/2024	Maximum Supply Voltage Updated from 3.2V to 3.3V	KL

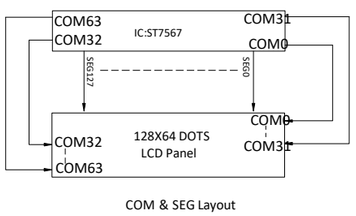
Mechanical Drawing



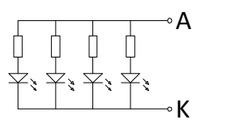




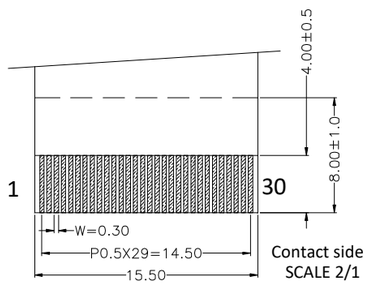
VIEWING DIRECTION

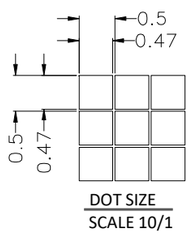


COM & SEG Layout



BACKLIGHT LECIRCUIT(4PCS LED)
COLOR: WHITE





PIN No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Symbol	CSB	RSTB	A0	R/W	E	DB0	DB1	DB2	DB3	DB4	DB5	DB6	DB7	VDD	VSS

PIN No.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Symbol	NC	NC	NC	NC	V0	XV0	VG	NC	NC	NC	NC	NC	C86	PSB	NC

Product Description: 128x64 Graphic LCD

1. Driver IC: ST7567
2. Driving Mode: 1/65 Duty, 1/9 Bias
3. Interface: 8- bit 6800/8080 parallel, 4-Line SPI
4. Power Requirement: 3.0V LCD, 3.5V/34mA Backlight
5. Optical Features: FSTN (+) Transflective, 6:00 View, White Backlight
6. Recommended FFC Connector: 30pin 0.5mm pitch; Ex. Molex 52892-3095

Standard Tolerance:
(Unless otherwise specified)
Linear: ±0.3mm

NEWHAVEN DISPLAY INTERNATIONAL

Drawing/Part Number:	Revision:
NHD-C12864WO-B1TFH#-M	2A

Unless otherwise specified:
• Dimensions are in Millimeters
• Third Angle Projection

Drawn By:	Approved By:
K. Lewis	K. Lewis
Drawn Date: 07/26/2023	Approved Date: 07/26/2023

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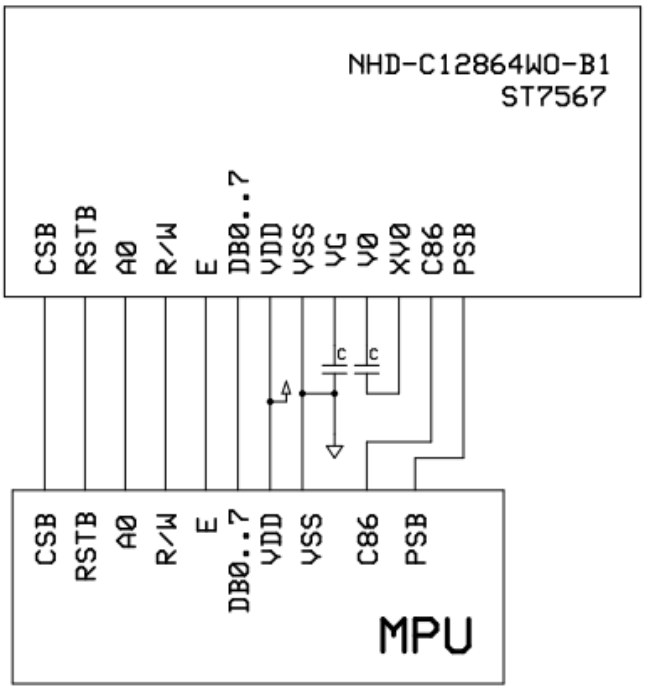
Pin Description

LCD:

Pin No.	Symbol	External Connection	Function Description
1	CSB	MPU	Active LOW Chip Select
2	RSTB	MPU	Active LOW RESET signal
3	A0	MPU	Register select signal. A0 = 0: instruction; A0 = 1: data
4	R/W /WR	MPU	6800 Mode: Read/Write select signal. R/W=1: Read R/W: =0: Write 8080 Mode: Active LOW Write Signal Serial Mode: +3V (*HIGH*)
5	E /RD	MPU	6800 Mode: Active HIGH Enable Signal 8080 Mode: Active LOW Read Signal Serial Mode: +3V (*HIGH*)
6-13	DB0~DB7	MPU	In parallel interface: 8-bit-directional data bus In serial interface: DB0~DB5 = VDD or NC D6 = Serial clock D7 = Serial data
14	VDD	Power Supply	Supply Voltage for logic
15	VSS	Power Supply	Ground
16~19	NC	-	No Connect
20	V0	Power Supply	Connect 0.1µF - 1µF Capacitor to XV0 (pin 21)
21	XV0	Power Supply	Connect 0.1µF - 1µF Capacitor to V0 (pin 20)
22	VG	Power Supply	Connect 0.1µF - 1µF Capacitor to V_{SS}
23-27	NC	-	No Connect
28	C86	MPU	Select MPU interface pin. C86= H:6800; C86= L:8080
29	PSB	MPU	Parallel/Serial select. PS= H: Parallel; PS= L: Serial
30	NC	-	No connect

Recommended LCD connector: 0.5mm Pitch, 30pin FFC connector. Molex p/n: 52892-3095

Wiring Diagram



Electrical Characteristics

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
Operating Temperature Range	T_{OP}	Absolute Max	-20	-	+70	°C
Storage Temperature Range	T_{ST}	Absolute Max	-30	-	+80	°C
Supply Voltage	V_{DD}	-	2.8	3.0	3.3	V
Supply Current	I_{DD}	Ta=25°C, V_{DD}=3.0V	-	0.7	1.0	mA
Supply for LCD (contrast)	V_{DD}-V0	Ta=25°C	9.3	9.5	9.7	V
"H" Level input	V_{IH}	-	0.7*V_{DD}	-	V_{DD}	V
"L" Level input	V_{IL}	-	V_{SS}	-	0.3*V_{DD}	V
"H" Level output	V_{OH}	-	0.8*V_{DD}	-	V_{DD}	V
"L" Level output	V_{OL}	-	V_{SS}	-	0.2*V_{DD}	V
Backlight Supply Voltage	V_{LED}	I_{LED} = 34mA	3.2	3.5	3.8	V
Backlight Supply Current	I_{LED}	-	20	34	80	mA
Backlight Lifetime	-	I_{LED} = 34mA T_{OP} = 25°C	-	50,000	-	Hrs.

*The LED of the backlight is driven by current drain; drive voltage is for reference only. Drive voltage must be selected to ensure backlight current drain is below MAX level stated.

Optical Characteristics

Item		Symbol	Condition	Min.	Typ.	Max.	Unit
Optimal Viewing Angles	Top	φY+	Cr ≥ 2	-	30	-	o
	Bottom	φY-		-	60	-	o
	Left	θX-		-	45	-	o
	Right	θX+		-	45	-	o
Contrast Ratio		Cr	-	-	5	-	-
Response Time	Rise	Tr	-	-	200	300	ms
	Fall	Tf	-	-	250	350	ms

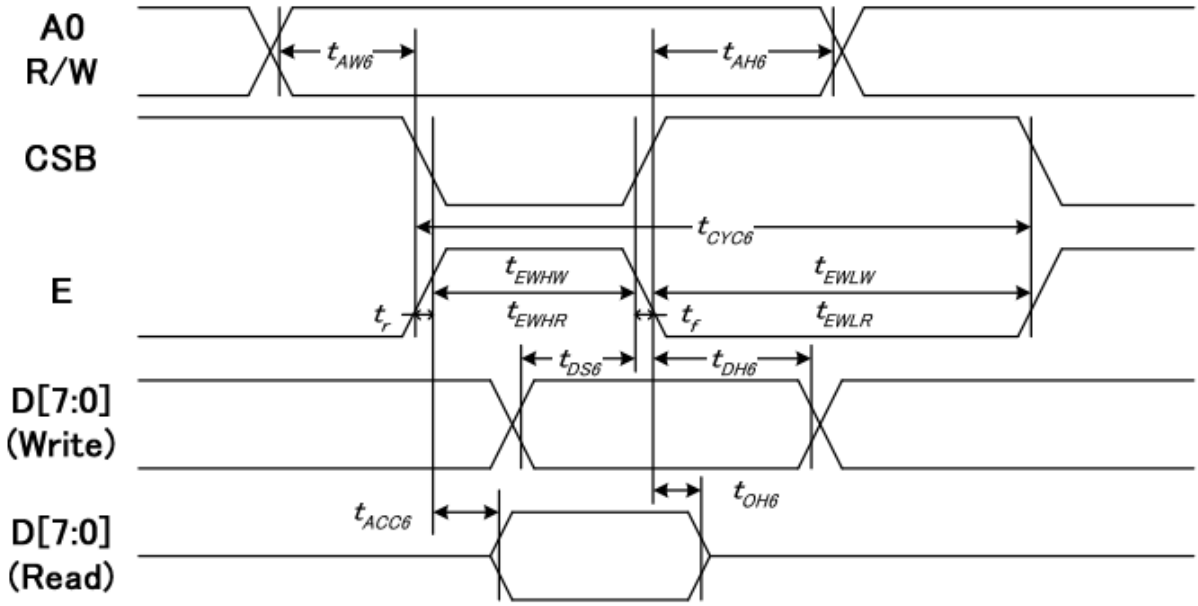
Controller Information

Built-in ST7567 Controller: https://support.newhavendisplay.com/hc/en-us/articles/4414899443607-ST7567

Timing Characteristics

6800 Parallel

System Bus Timing for 6800 Series MPU

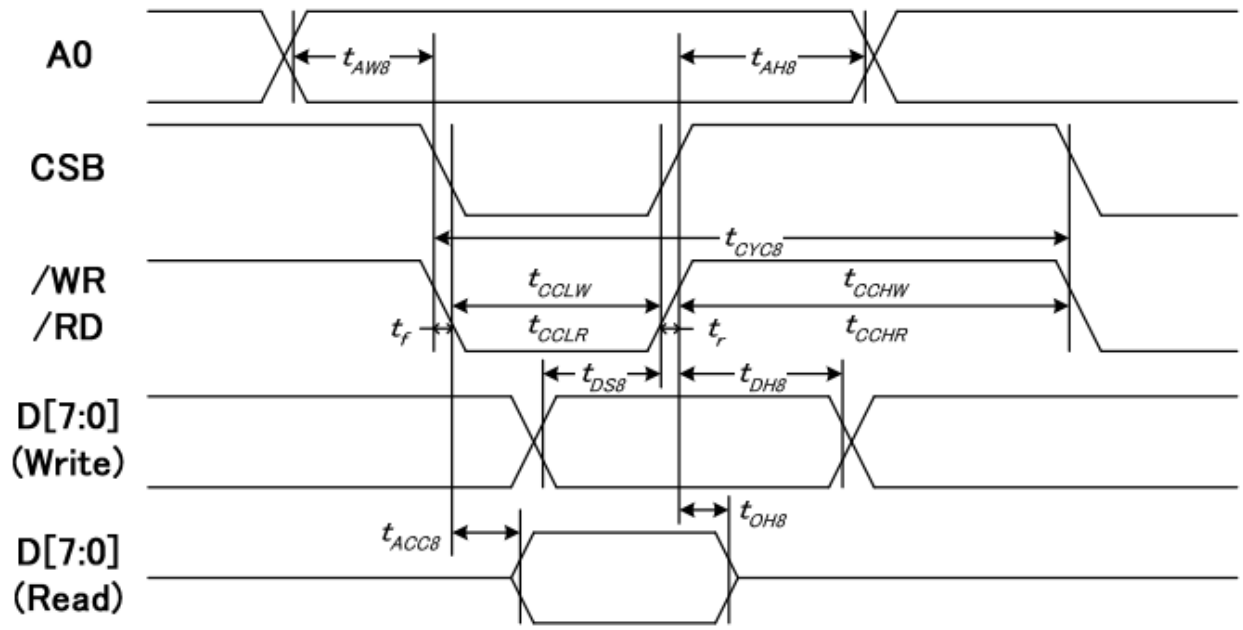


(VDD1 = 3.3V , Ta =25°C)

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Address setup time	A0	tAW6		0	—	ns
Address hold time		tAH6		10	—	
System cycle time	E	tCYC6		240	—	
Enable L pulse width (WRITE)		tEWLW		80	—	
Enable H pulse width (WRITE)		tEWHW		80	—	
Enable L pulse width (READ)		tEWLR		80	—	
Enable H pulse width (READ)		tEWHR		140		
Write data setup time	D[7:0]	tDS6		40	—	
Write data hold time		tDH6		10	—	
Read data access time		tACC6	CL = 16 pF	—	70	
Read data output disable time		tOH6	CL = 16 pF	5	50	

8080 Parallel

System Bus Timing for 8080 Series MPU

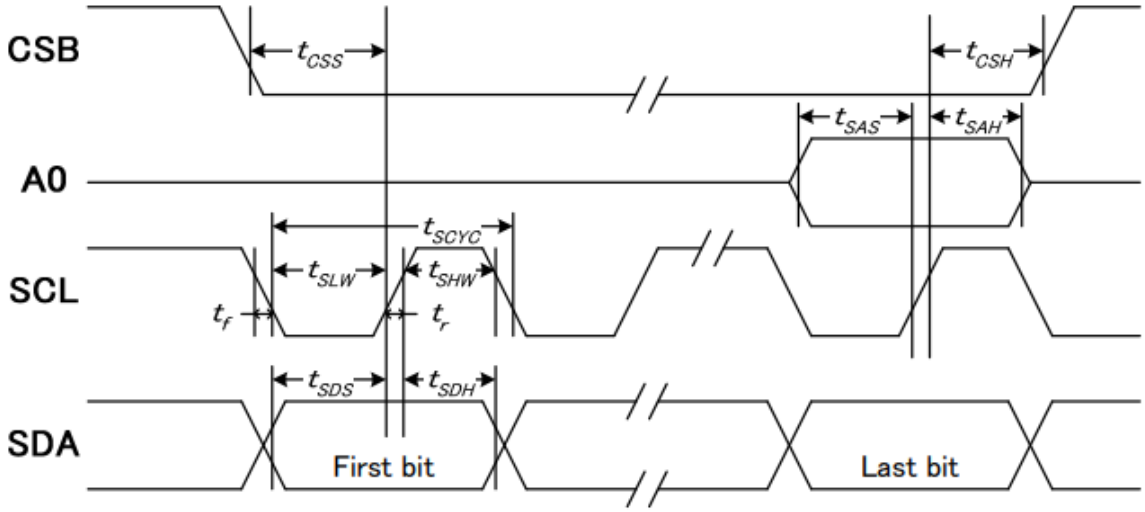


(VDD1 = 3.3V , Ta =25°C)

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Address setup time	A0	tAW8		0	—	ns
Address hold time		tAH8		10	—	
System cycle time	/WR	tCYC8		240	—	
/WR L pulse width (WRITE)		tCCLW		80	—	
/WR H pulse width (WRITE)		tCCHW		80	—	
/RD L pulse width (READ)	RD	tCCLR		140	—	
/RD H pulse width (READ)		tCCHR		80		
WRITE Data setup time	D[7:0]	tDS8		40	—	
WRITE Data hold time		tDH8		20	—	
READ access time		tACC8	CL = 16 pF	—	70	
READ Output disable time		tOH8	CL = 16 pF	5	50	

4-Line SPI

System Bus Timing for 4-Line Serial Interface



(VDD1 = 3.3V , Ta =25°C)

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Serial clock period	SCLK	tSCYC		50	—	ns
SCLK “H” pulse width		tSHW		25	—	
SCLK “L” pulse width		tSLW		25	—	
Address setup time	A0	tSAS		20	—	
Address hold time		tSAH		10	—	
Data setup time	SDA	tSDS		20	—	
Data hold time		tSDH		10	—	
CSB-SCLK time	CSB	tCSS		20	—	
CSB-SCLK time		tCSH		40	—	

Table of Commands

INSTRUCTION	A0	R/W (RWR)	COMMAND BYTE D7	D6	D5	D4	D3	D2	D1	D0	DESCRIPTION
(1) Display ON/OFF	0	0	1	0	1	0	1	1	1	D	D=1, display ON D=0, display OFF
(2) Set Start Line	0	0	0	1	S5	S4	S3	S2	S1	S0	Set display start line
(3) Set Page Address	0	0	1	0	1	1	Y3	Y2	Y1	Y0	Set page address
(4) Set Column Address	0	0	0	0	0	1	X7	X6	X5	X4	Set column address (MSB)
	0	0	0	0	0	0	X3	X2	X1	X0	Set column address (LSB)
(5) Read Status	0	1	0	MX	D	RST	0	0	0	0	Read IC Status
(6) Write Data	1	0	D7	D6	D5	D4	D3	D2	D1	D0	Write display data to RAM
(7) Read Data	1	1	D7	D6	D5	D4	D3	D2	D1	D0	Read display data from RAM
(8) SEG Direction	0	0	1	0	1	0	0	0	0	MX	Set scan direction of SEG MX=1, reverse direction MX=0, normal direction
(9) Inverse Display	0	0	1	0	1	0	0	1	1	INV	INV =1, inverse display INV =0, normal display
(10) All Pixel ON	0	0	1	0	1	0	0	1	0	AP	AP=1, set all pixel ON AP=0, normal display
(11) Bias Select	0	0	1	0	1	0	0	0	1	BS	Select bias setting 0=1/9; 1=1/7 (at 1/65 duty)
(12) Read-modify-Write	0	0	1	1	1	0	0	0	0	0	Column address increment: Read:+0 , Write:+1
(13) END	0	0	1	1	1	0	1	1	1	0	Exit Read-modify-Write mode
(14) RESET	0	0	1	1	1	0	0	0	1	0	Software reset
(15) COM Direction	0	0	1	1	0	0	MY	-	-	-	Set output direction of COM MY=1, reverse direction MY=0, normal direction
(16) Power Control	0	0	0	0	1	0	1	VB	VR	VF	Control built-in power circuit ON/OFF
(17) Regulation Ratio	0	0	0	0	1	0	0	RR2	RR1	RR0	Select regulation resistor ratio
(18) Set EV	0	0	1	0	0	0	0	0	0	1	Double command!! Set electronic volume (EV) level
	0	0	0	0	EV5	EV4	EV3	EV2	EV1	EV0	
(19) Set Booster	0	0	1	1	1	1	1	0	0	0	Double command!! Set booster level: BL=0: 4X BL=1: 5X
	0	0	0	0	0	0	0	0	0	BL	
(20) Power Save	0	0	Compound Command								Display OFF + All Pixel ON
(21) NOP	0	0	1	1	1	0	0	0	1	1	No operation
(22) Test	0	0	1	1	1	1	1	1	1	-	Do NOT use. Reserved for testing.

Note: Symbol "-" means this bit can be "H" or "L".

Quality Information

Test Item	Content of Test	Test Condition	Note
High Temperature storage	Endurance test applying the high storage temperature for a long time.	+80°C , 200hrs	2
Low Temperature storage	Endurance test applying the low storage temperature for a long time.	-30°C , 200hrs	1,2
High Temperature Operation	Endurance test applying the electric stress (voltage & current) and the high thermal stress for a long time.	+70°C 200hrs	2
Low Temperature Operation	Endurance test applying the electric stress (voltage & current) and the low thermal stress for a long time.	-20°C , 200hrs	1,2
High Temperature / Humidity Operation	Endurance test applying the electric stress (voltage & current) and the high thermal with high humidity stress for a long time.	+60°C , 90% RH , 96hrs	1,2
Thermal Shock resistance	Endurance test applying the electric stress (voltage & current) during a cycle of low and high thermal stress.	-20°C,30min -> 25°C,5min -> 70°C,30min = 1 cycle 10 cycles	
Vibration test	Endurance test applying vibration to simulate transportation and use.	10-55Hz , 1.5mm amplitude. 2g Acceleration. 60 sec in each of 3 directions X,Y,Z For 30 minutes	3
Static electricity test	Endurance test applying electric static discharge.	Air: ±8KV, Contact: ±4KV	4

Note 1: No condensation to be observed.
Note 2: Conducted after 4 hours of storage at 25°C, 0%RH.
Note 3: Test performed on product itself, not inside a container.